CARD A01 | STRUCTURE → NAME

A01 - FOUNDATION - C_6H_{14}

1. NUMBERED PARENT



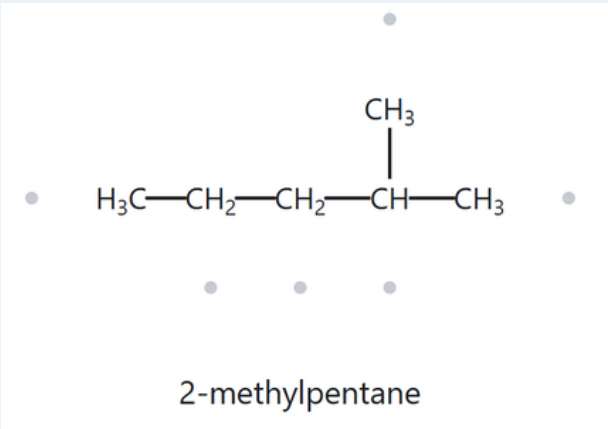
2. NAMING TRAIL

Parent: longest chain = 5 C → pentane.
Numbering: one end gives locant 4, the other gives 2. 2 < 4, so number from the end nearer the branch.
Substituents: methyl on C2.

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



Returned name:

2-methylpentane

✓ **Match**

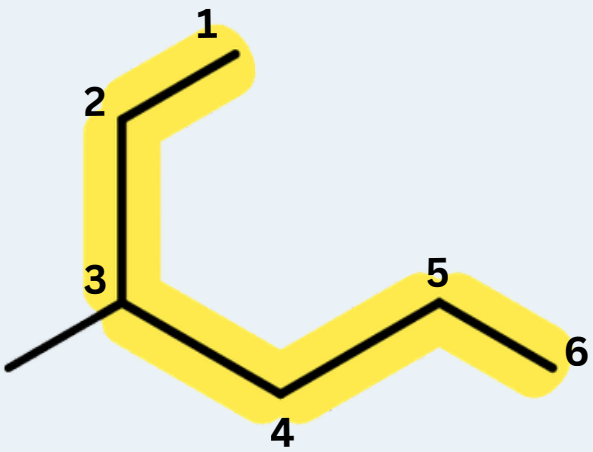
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A02 | STRUCTURE → NAME

A02 - FOUNDATION - C_7H_{16}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: the horizontal path is only 5 C. Tracing from the end of the ethyl branch through the branch point and along the chain gives 6 C → hexane.

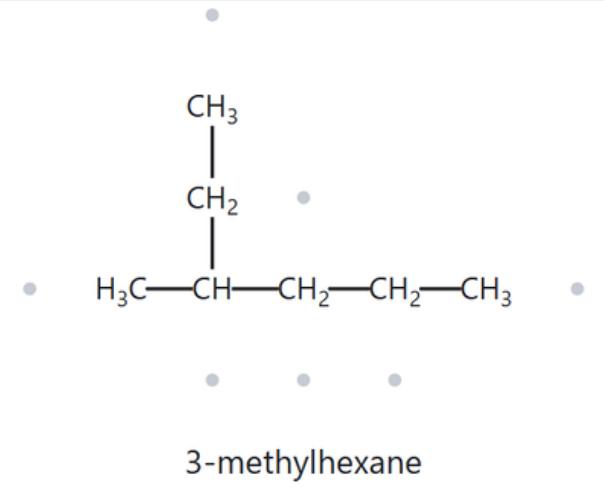
Numbering: the branch is at 3 from one end and 4 from the other. 3 < 4, so use the end that gives 3.

Substituents: methyl on C3.

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



Returned name:

3-methylhexane

✓ **Match**

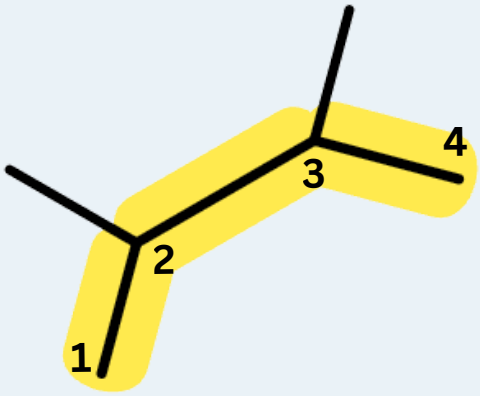
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A03 | STRUCTURE → NAME

A03 - FOUNDATION - C_6H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C → butane (the two arms off each central carbon are 1 C, so no 5 C path exists).

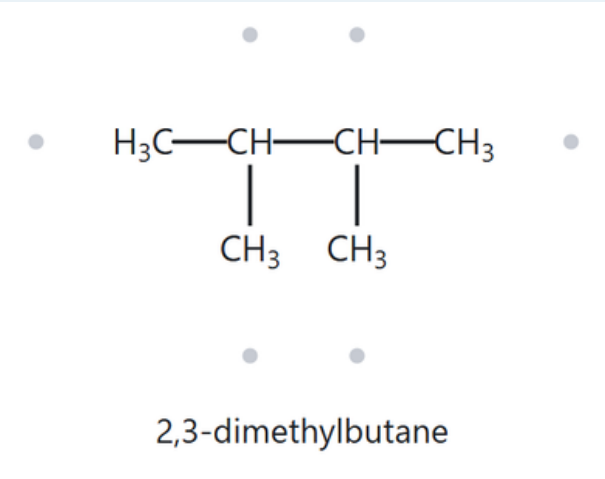
Numbering: both directions give {2,3}, so it is a tie and either way is fine.

Substituents: methyl on C2 and C3 → di- prefix.

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



Returned name:

2,3-dimethylbutane

✓ **Match**

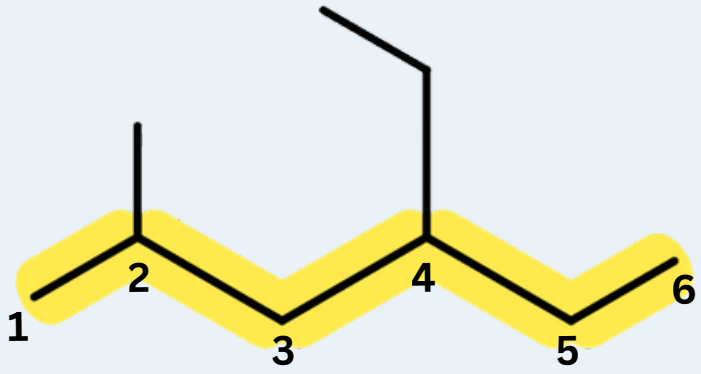
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A04 | STRUCTURE → NAME

A04 - DEVELOPING - C_9H_{20}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C → hexane. Other 6 C paths tie but carry the same substituents.

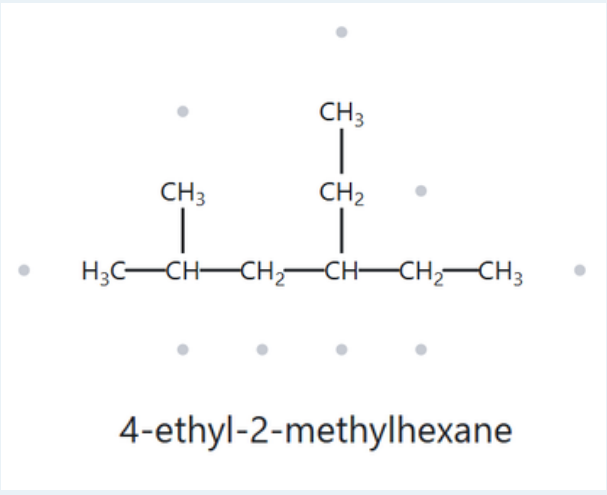
Numbering: one direction gives {2,4} and the other gives {3,5}. 2 < 3 at the first point of difference, so use {2,4}.

Substituents: ethyl on C4, methyl on C2. Alphabetical order puts ethyl before methyl, regardless of locants.

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



Returned name:

4-ethyl-2-methylhexane

✓ **Match**

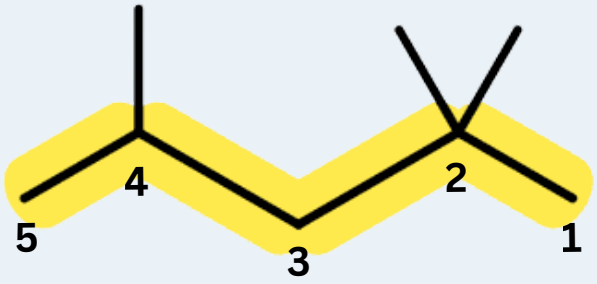
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A05 | STRUCTURE → NAME

A05 - DEVELOPING - C_8H_{18}

1. NUMBERED PARENT



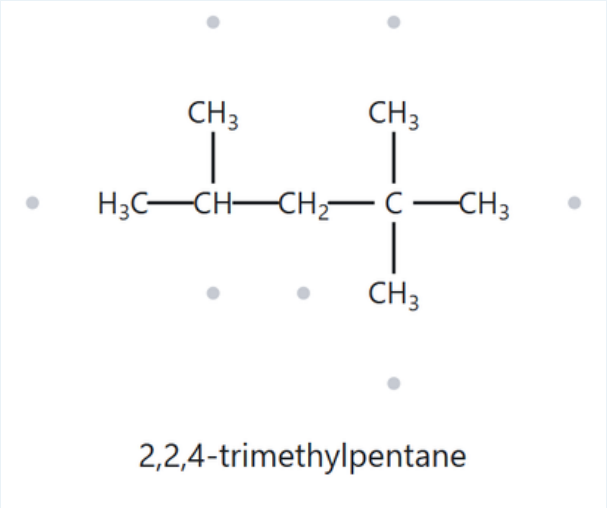
2. NAMING TRAIL

Parent: longest chain = 5 C → pentane.
Numbering: one direction gives {2,2,4} and the other gives {2,4,4}. The first terms tie at 2, and the second terms are 2 < 4, so use {2,2,4}.
Substituents: methyl on C2, C2 and C4 → tri- prefix.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



Returned name:
2,2,4-trimethylpentane
✓ **Match**

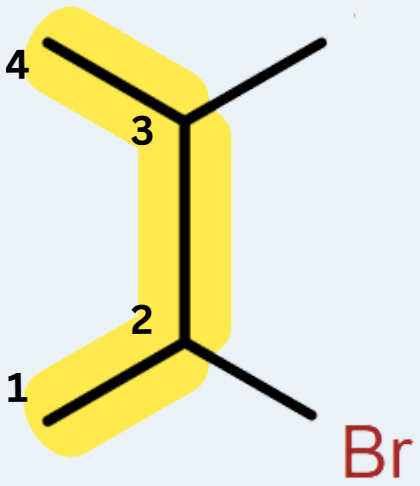
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A06 | STRUCTURE → NAME

A06 - DEVELOPING - $C_5H_{11}Br$

1. NUMBERED PARENT



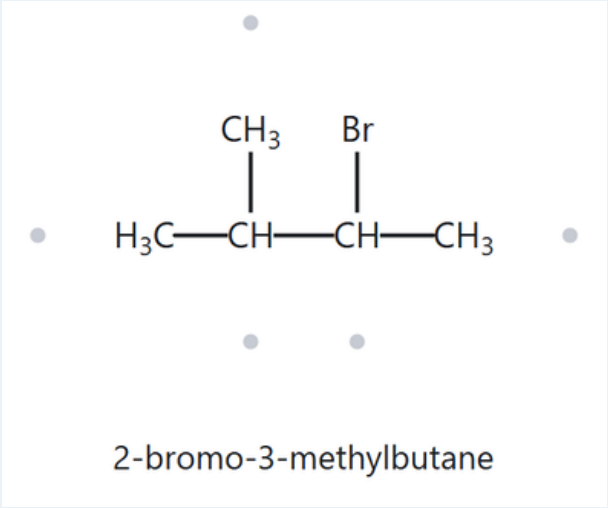
2. NAMING TRAIL

Parent: longest chain = 4 C → butane.
Numbering: both directions give {2,3}, a tie. Bromo comes before methyl alphabetically, so bromo gets the lower number (C2).
Substituents: bromo on C2, methyl on C3.

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



Returned name:
2-bromo-3-methylbutane

✓ **Match**

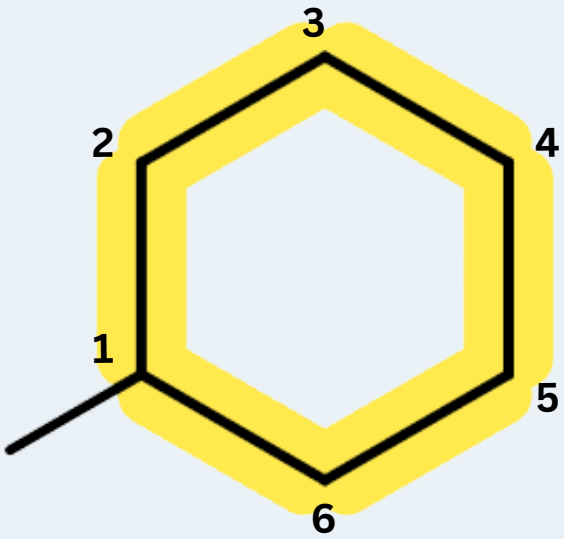
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A07 | STRUCTURE → NAME

A07 - DEVELOPING - C_7H_{14}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: ring of 6 C → cyclohexane (the ring is the parent).

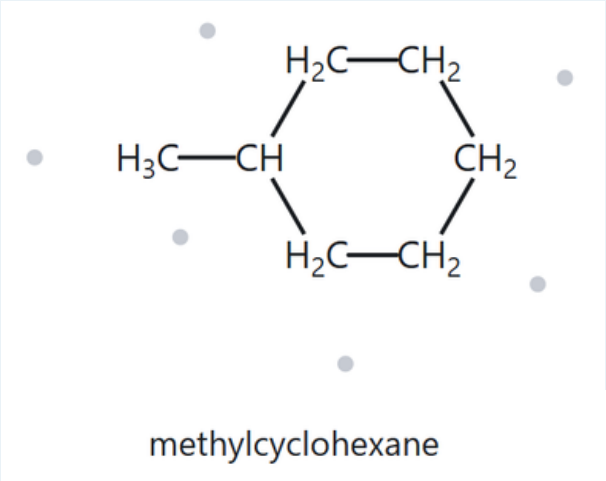
Numbering: only one substituent, so any ring carbon can be C1 and no locant is written.

Substituents: methyl (no locant).

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



Returned name:

methylcyclohexane

✓ **Match**

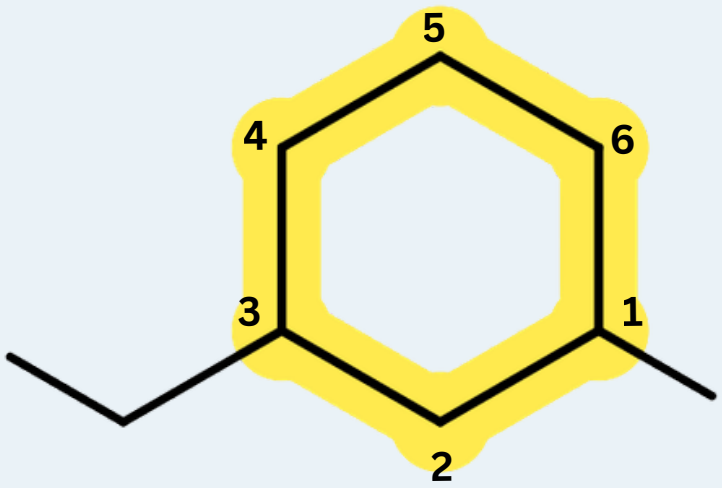
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A08 | STRUCTURE → NAME

A08 - CHALLENGE - C_9H_{18}

1. NUMBERED PARENT



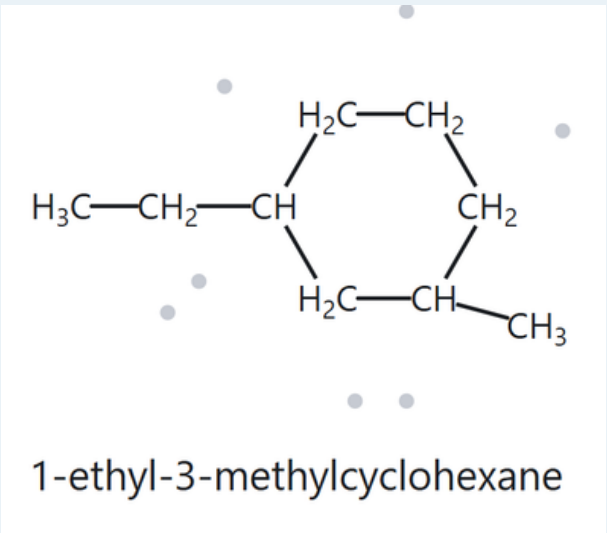
2. NAMING TRAIL

Parent: ring of 6 C → cyclohexane.
Numbering: the substituents are on carbons 1,3 whichever way round you go, so the locant sets tie. Ethyl comes before methyl alphabetically, so ethyl gets C1.
Substituents: ethyl on C1, methyl on C3.

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



Returned name:
1-ethyl-3-methylcyclohexane

✓ **Match**

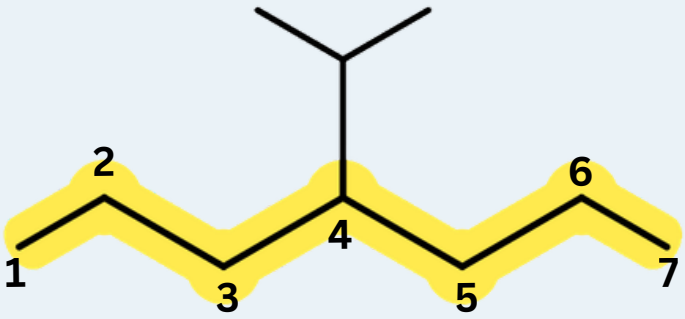
VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A09 | STRUCTURE → NAME

A09 - CHALLENGE - $C_{10}H_{22}$

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: the longest chain is the 7 C zigzag. A path through the branch gives only 5 C, so the branch is a substituent.

Numbering: the branch is on the middle carbon, so both directions give 4.

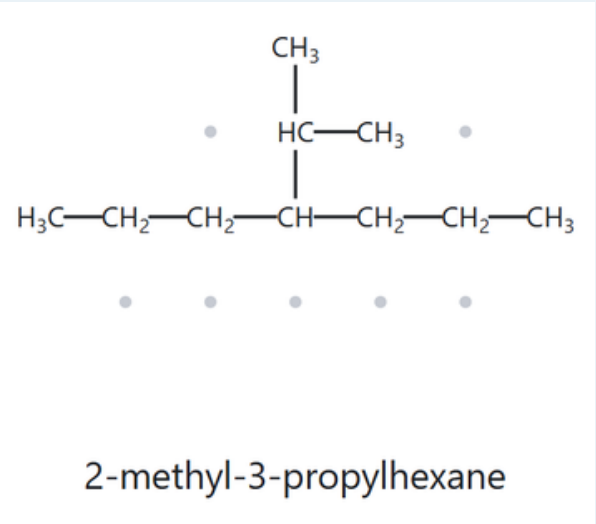
Substituents: isopropyl on C4.

Equivalent accepted forms are 4-(1-methylethyl)heptane and 4-(propan-2-yl)heptane.

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



Returned name:

2-methyl-3-propylhexane

✗ **Match**

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

CARD A10 | STRUCTURE → NAME

A10 - CHALLENGE - C_9H_{18}

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: the ring (3 C) is the parent under this activity's rule, even though the chain has more carbons → cyclopropane.

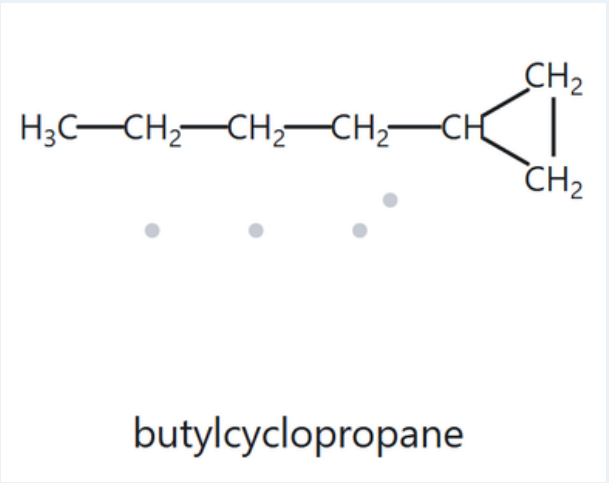
Numbering: only one substituent, so no locant is needed.

Substituents: butyl, a 4 C unbranched chain (no locant).

3. MY NAME

butylcyclopropane

4. THE ALKANE CHECK



Returned name:

butylcyclopropane

✓ **Match**

VERIFICATION STRIP

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order ☑ punctuation ☑ formula counted

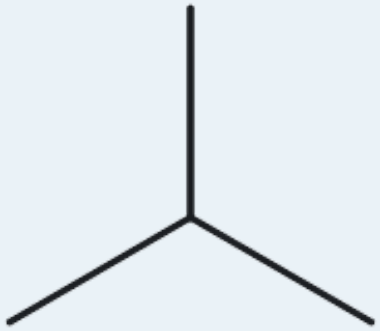
CARD B01 | NAME → STRUCTURE

B01 - FOUNDATION - 2-methylpropane

1. NAME DECODED

Parent: propane → a 3-carbon chain.
Substituents: methyl on C2.
Expected formula: 3 + 1 = 4 C; open chain → C4H10.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C4H10

4. BACK-CHECK

Returned name:

2-methylpropane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

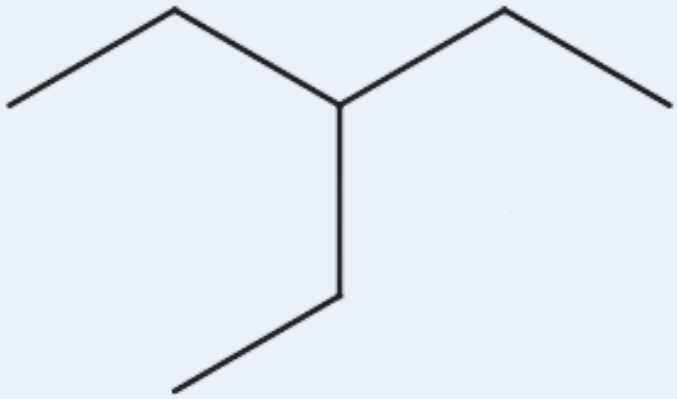
CARD B02 | NAME → STRUCTURE

B02 - FOUNDATION - 3-ethylpentane

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected formula: 5 + 2 = 7 C; open chain → C7H16.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C7H16

4. BACK-CHECK

Returned name:

3-ethylpentane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

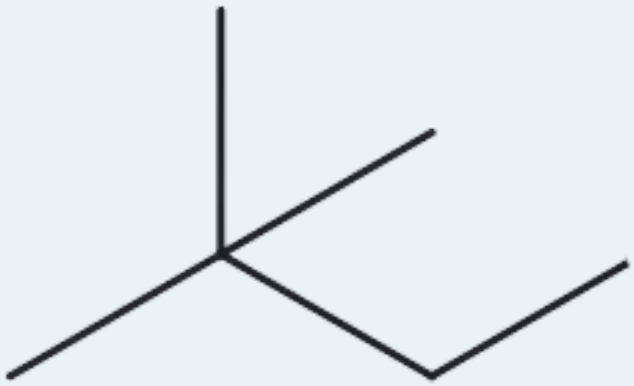
CARD B03 | NAME → STRUCTURE

B03 - FOUNDATION - 2,2-dimethylbutane

1. NAME DECODED

Parent: butane → a 4-carbon chain.
Substituents: two methyls, both on C2.
Expected formula: 4 + 2 = 6 C; open chain → C6H14.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C6H14

4. BACK-CHECK

Returned name:

2,2-dimethylbutane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

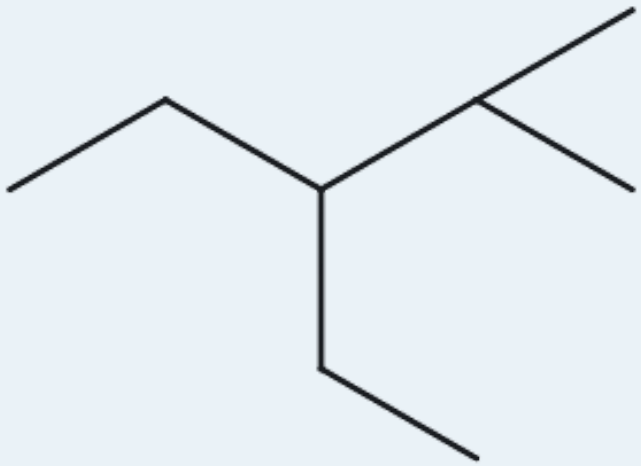
CARD B04 | NAME → STRUCTURE

B04 - DEVELOPING - 3-ethyl-2-methylpentane

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: methyl on C2; ethyl on C3.
Expected formula: 5 + 1 + 2 = 8 C; open chain → C8H18.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C8H18

4. BACK-CHECK

Returned name:

3-ethyl-2-methylpentane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

CARD B05 | NAME → STRUCTURE

B05 - DEVELOPING - 1-chloro-2-methylpropane

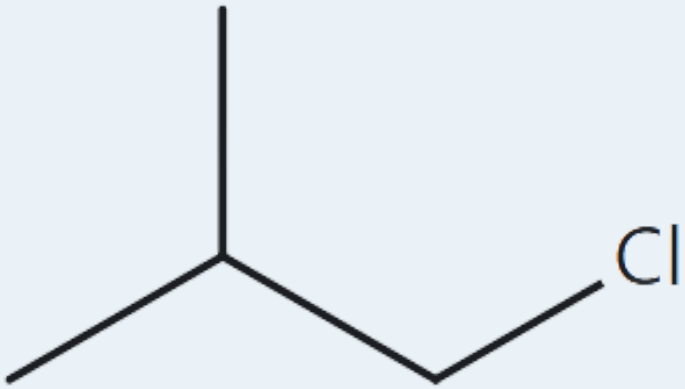
1. NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: chloro on C1; methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C4H10; Cl replaces one H → C4H9Cl.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C4H9Cl

4. BACK-CHECK

Returned name:

1-chloro-2-methylpropane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

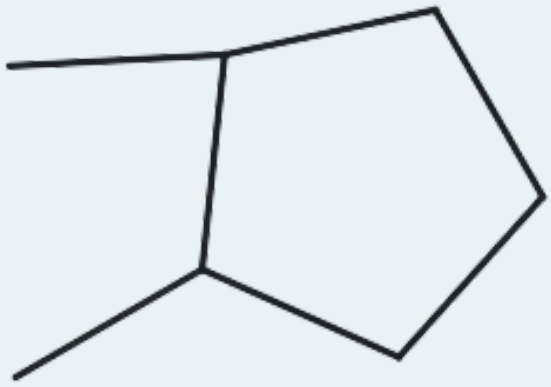
CARD B06 | NAME → STRUCTURE

B06 - DEVELOPING - 1,2-dimethylcyclopentane

1. NAME DECODED

Parent: cyclopentane → a 5-carbon ring.
Substituents: methyl on C1; methyl on C2 (adjacent carbons).
Expected formula: 5 + 2 = 7 C; one ring → C7H14.

2 BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C7H14

4. BACK-CHECK

Returned name:

1,2-dimethylcyclopentane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

CARD B07 | NAME → STRUCTURE

B07 - DEVELOPING - 4-ethyl-3,3-dimethylheptane

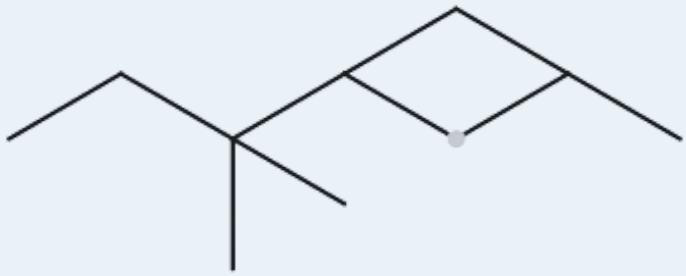
1. NAME DECODED

Parent: heptane → a 7-carbon chain.

Substituents: two methyls on C3; ethyl on C4.

Expected formula: 7 + 2 + 2 = 11 C; open chain → C11H24.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C11H24

4. BACK-CHECK

Returned name:

4-ethyl-3,3-dimethylheptane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

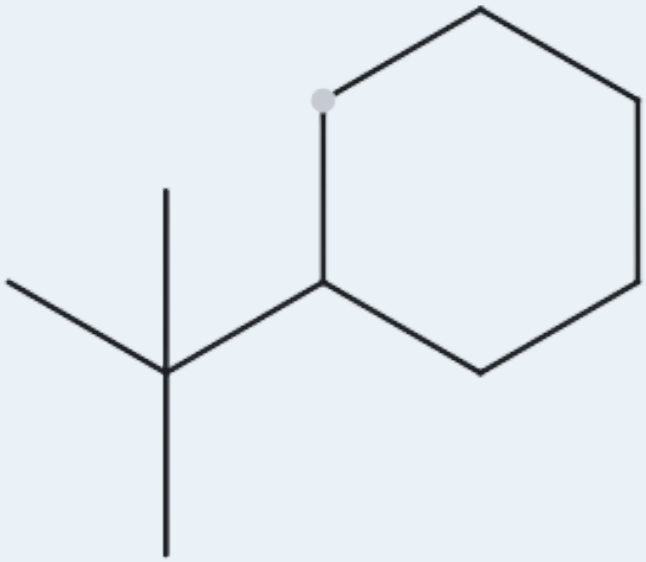
CARD B08 | NAME → STRUCTURE

B08 - CHALLENGE - tert-butylcyclohexane

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: tert-butyl, –C(CH3)3, a 4-carbon group (one quaternary carbon with three methyls). One substituent, so no locant.
Expected formula: 6 + 4 = 10 C; one ring → C10H20.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C10H20

4. BACK-CHECK

Returned name:

(1,1-dimethylethyl)cyclohexane

✗ **Not same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

CARD B09 | NAME → STRUCTURE

B09 - CHALLENGE - 3-ethyl-1,1-dimethylcyclohexane

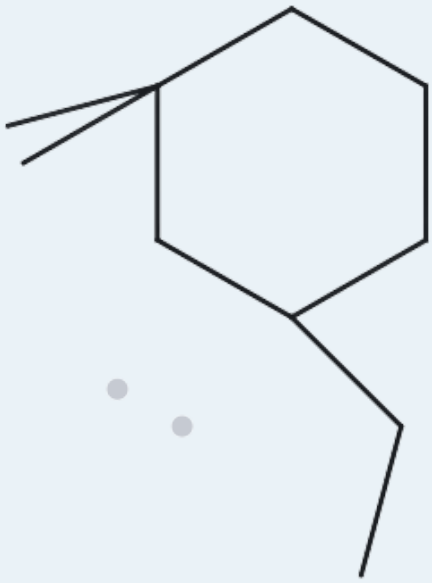
1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: two methyls on C1; ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → C10H20.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C10H20

4. BACK-CHECK

Returned name:

3-ethyl-1,1-dimethylcyclohexane

✓ **Same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded

CARD B10 | NAME → STRUCTURE

B10 - CHALLENGE - 2-ethylbutane

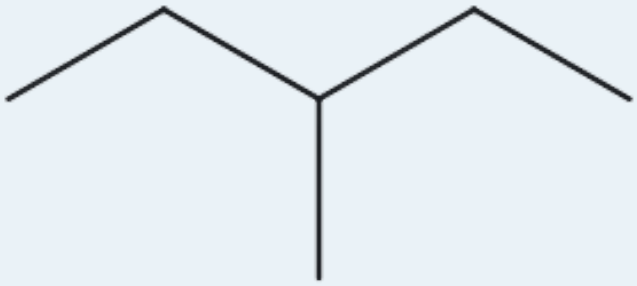
1. NAME DECODED

Parent: a 4-carbon chain.

Substituents: ethyl on C2.

Expected formula: 4 + 2 = 6 C; open chain → C6H14.

2 BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C6H14

4. BACK-CHECK

Returned name:

3-methylpentane

✗ **Not same as the given name**

VERIFICATION STRIP

☑ parent length matches ☑ substituent match ☑ formula agrees ☑ back-check recorded